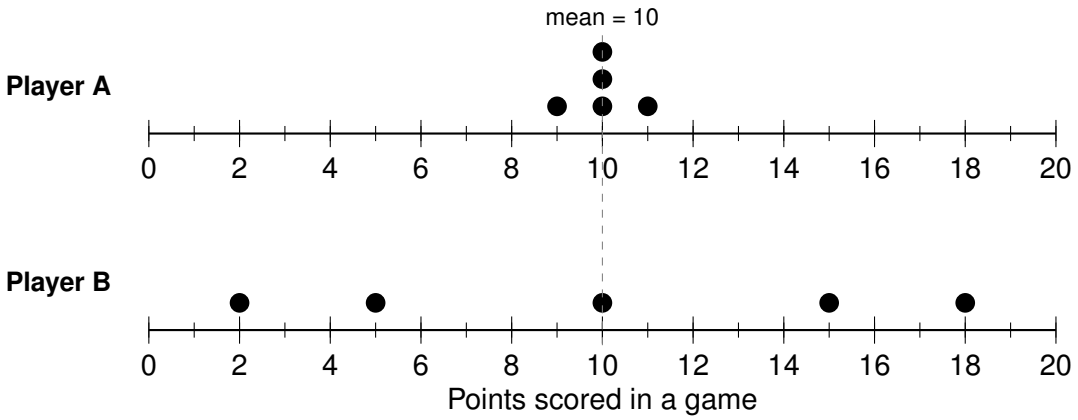
Standard Deviation

Name _____

Two basketball players both score **10 points** per game on average.



Player A's points are close to the mean. Player B's points are spread far from the mean. The **standard deviation** (SD) measures this.

Standard deviation is about how far the values usually are from the mean.
Small SD: values are close to the mean. **Large SD:** values are spread out.

How to find the standard deviation

1. Find the mean, $\bar{x}$. $\bar{x} = 50 \div 5 = 10$
2. Subtract the mean from each value.
3. Square each result.
4. Add the squares. $9 + 9 + 0 + 9 + 9 = 36$
5. Divide by $n - 1$. $36 \div 4 = 9$
   (This number is called the **variance**.)
6. Take the square root. $s = \sqrt{9} = \mathbf{3}$

Example: 7, 7, 10, 13, 13

x	$x - \bar{x}$	$(x - \bar{x})^2$
7	-3	9
7	-3	9
10	0	0
13	3	9
13	3	9
	Sum	36

The formula: $s = \sqrt{\dfrac{\text{sum of } (x - \bar{x})^2}{n - 1}}$ (n = how many values)

Why $n - 1$? Our data is usually a **sample**, a small part of a bigger group. A sample is usually a little less spread out than the whole group. Dividing by $n - 1$ makes s a little bigger to fix this.

Remember:

- The SD is never negative.
- The SD is 0 only when every value is the same.
- The SD has the same units as the data (points, minutes, inches).

Work with a partner. Compare answers when you finish each problem.

1. Find the standard deviation of 4, 6, 8, 10, 12.

x	$x - \bar{x}$	$(x - \bar{x})^2$
4		
6		
8		
10		
12		
	Sum	

Mean:

$s =$

2. Four quiz scores: 7, 3, 7, 7. Find the standard deviation.

x	$x - \bar{x}$	$(x - \bar{x})^2$
7		
3		
7		
7		
	Sum	

Mean:

$s =$

3. Look at the two players on page 1.

   Player A: 9, 10, 10, 11, 10
   Player B: 2, 18, 5, 15, 10

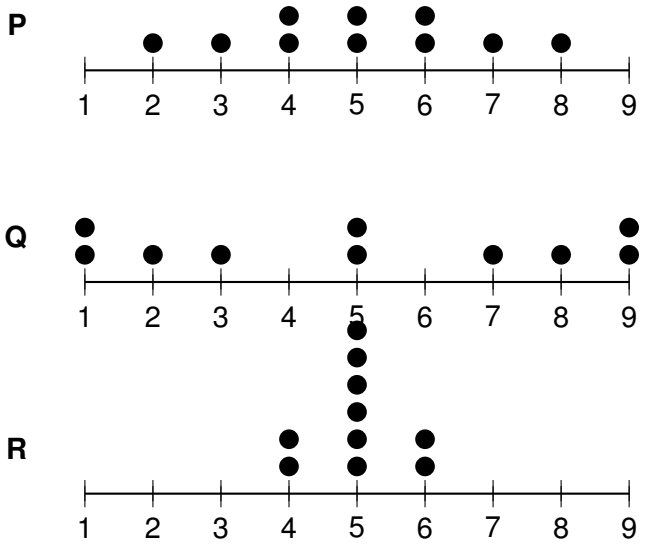
   (a) Find the mean of each player.

   (b) Without calculating, which player has the larger SD? Explain.

   (c) Find s for Player A.

   (d) Player B's SD is about 6.67. A coach needs a player who scores close to 10 points every game. Which player should the coach choose? Explain.

4. All three dot plots have a mean of 5. Put them in order from the **smallest** SD to the **largest** SD. Explain.



5. Think about what the SD means.

   (a) Can a standard deviation be negative? Explain.

   (b) Find the SD of 10, 10, 10, 10 without a formula. Explain.

   (c) Set B is 9, 10, 10, 11. Set C is 0, 10, 10, 20. Both have a mean of 10. Which set has the larger SD? Explain.